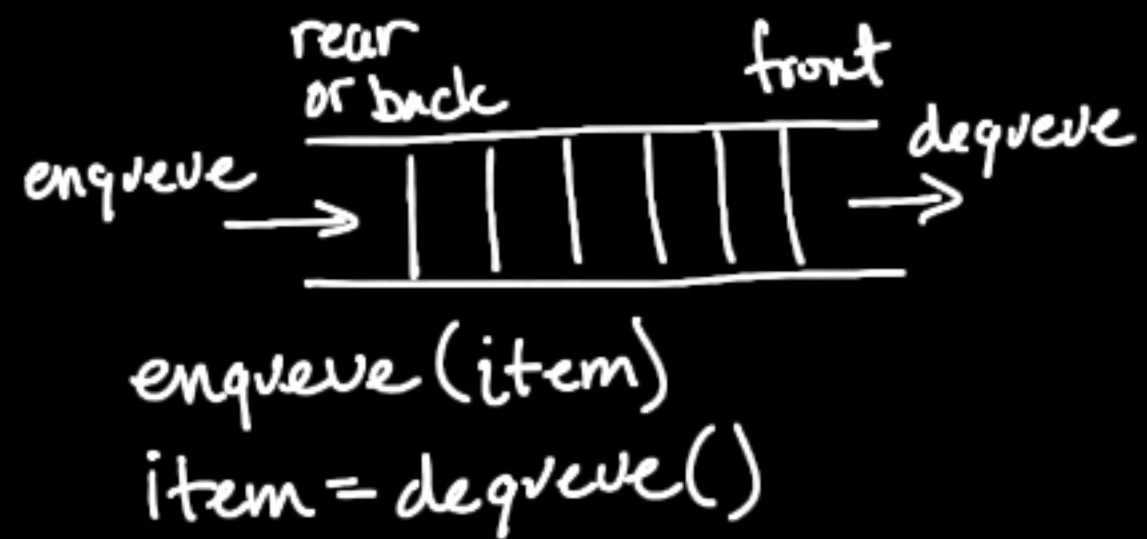
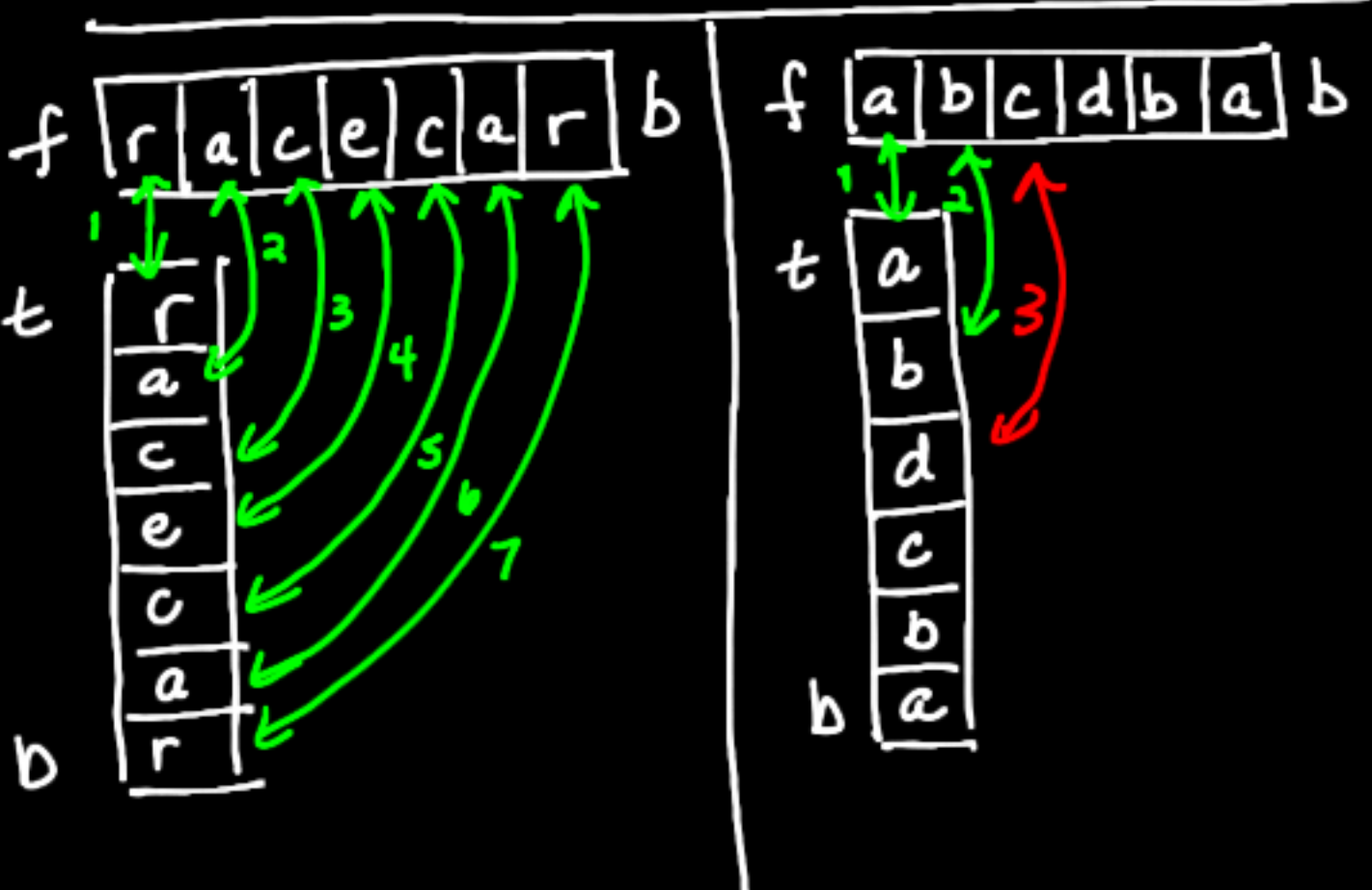


① Queue - FIFO

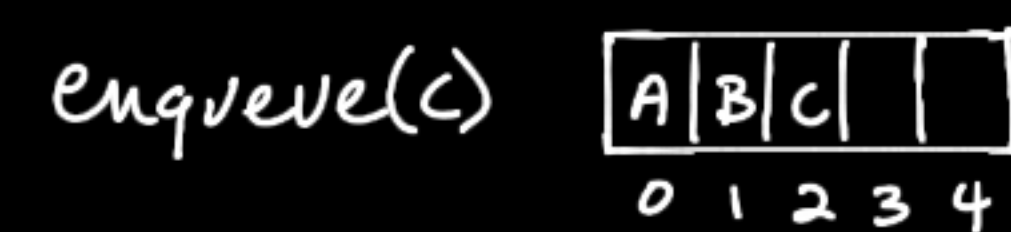
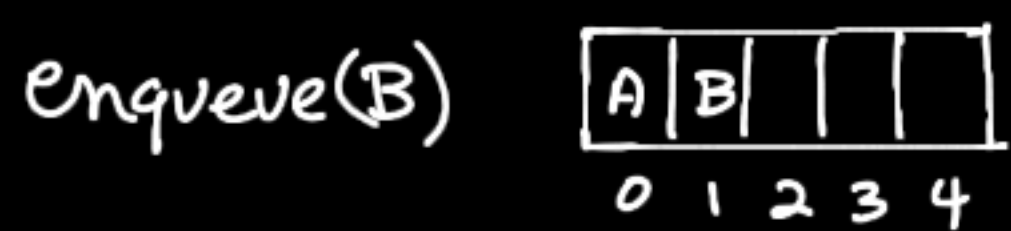


LIFO can reverse the order
FIFO can preserve the order

Example: recognizing palindromes
String: abcde
Queued: front [a|b|c|d|e] back
Stacked: top [e|d|c|b|a] base
qitem = dequeue()
sitem = pop()
qitem != sitem
∴ not a palindrome

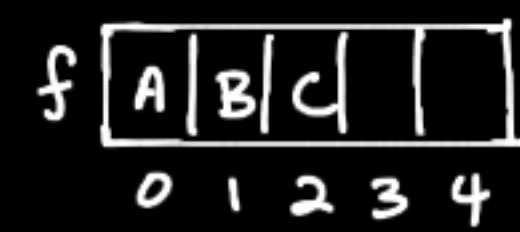


② Thought Experiment - Array-based Queue



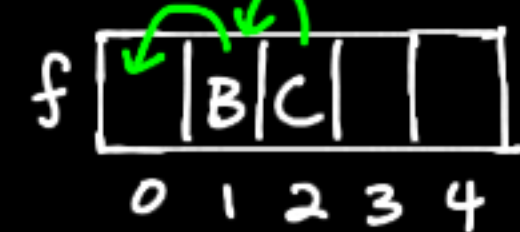
Keeping track

enqueue



b = b + 1

dequeue



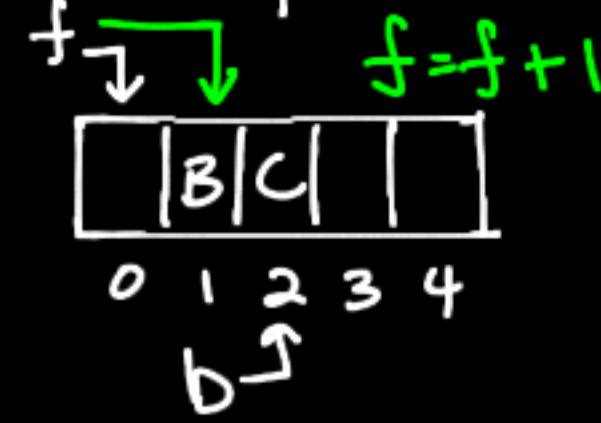
b = b - 1

+ Easy to implement
- Costly performance

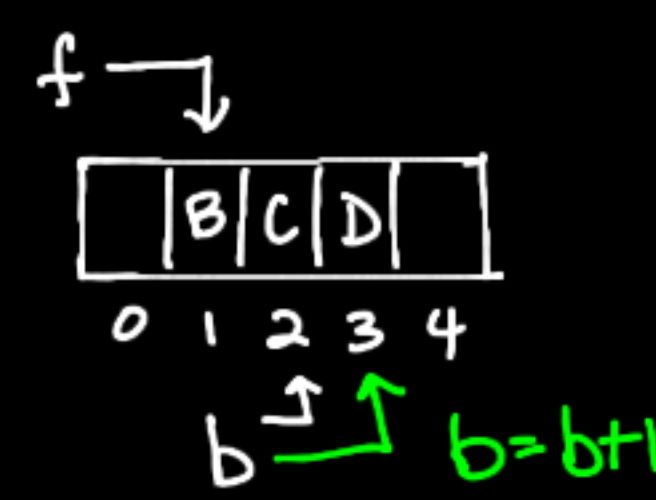
Can we do it without shifting?

③ Better - Adjust both front and back

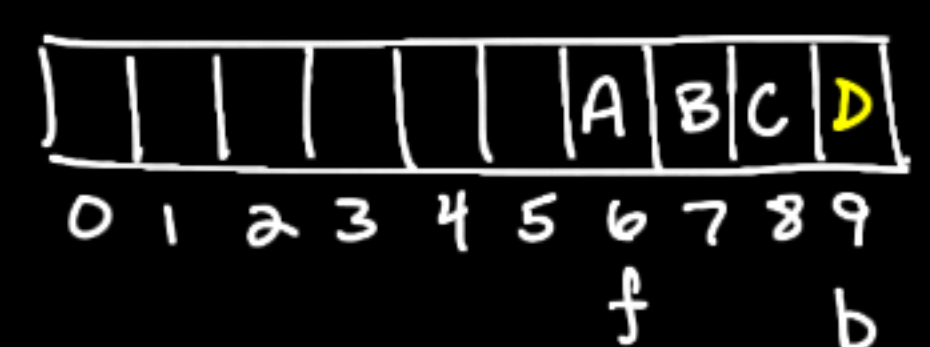
item = dequeue()



enqueue(D)

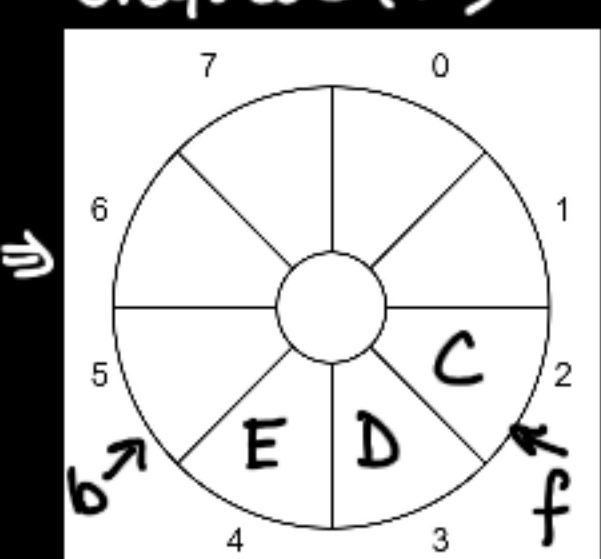
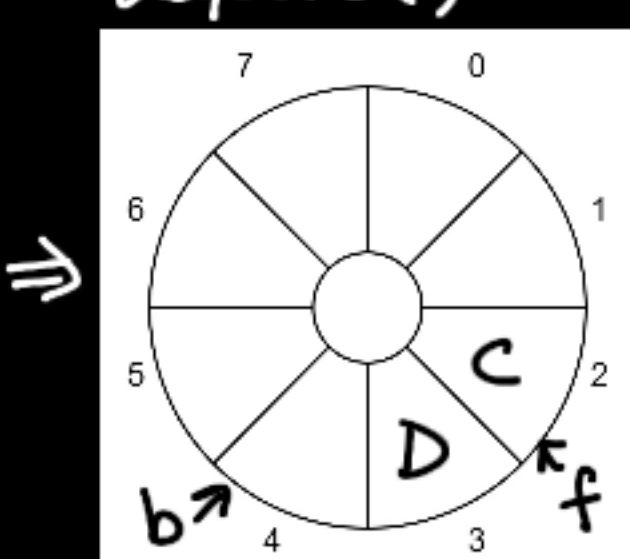
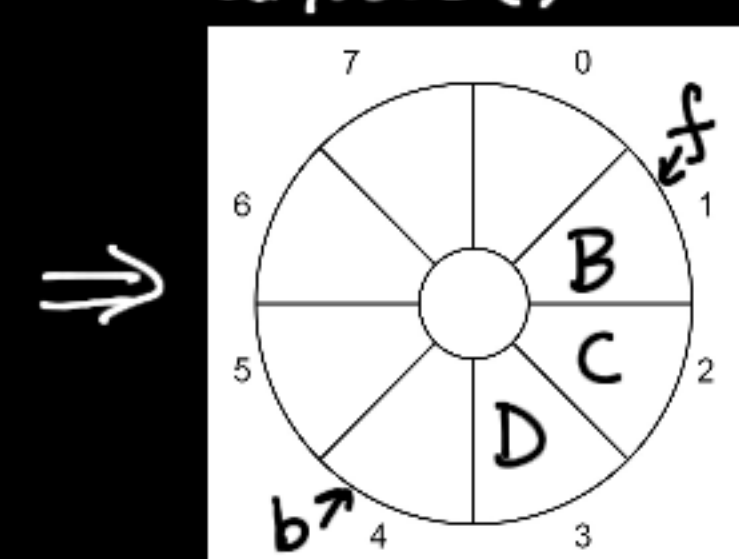
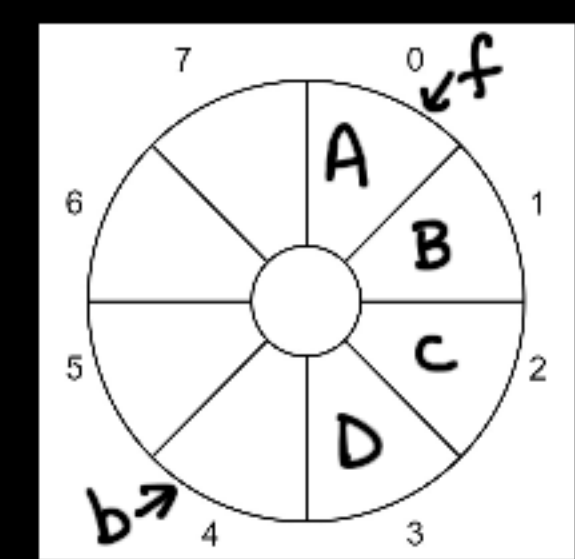


- 1) enqueue(D)
- 2) dequeue()
- 3) enqueue(E)
- 4) dequeue()
- 5) enqueue(F)
- 6) dequeue()
- 7) enqueue(G)
- 8) dequeue()

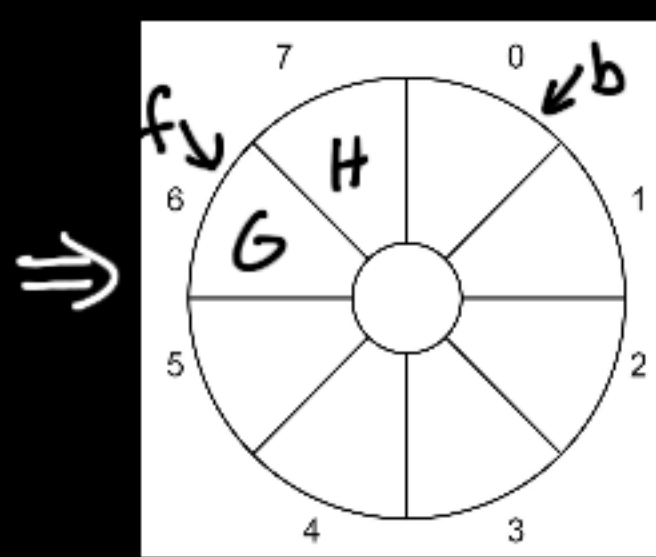
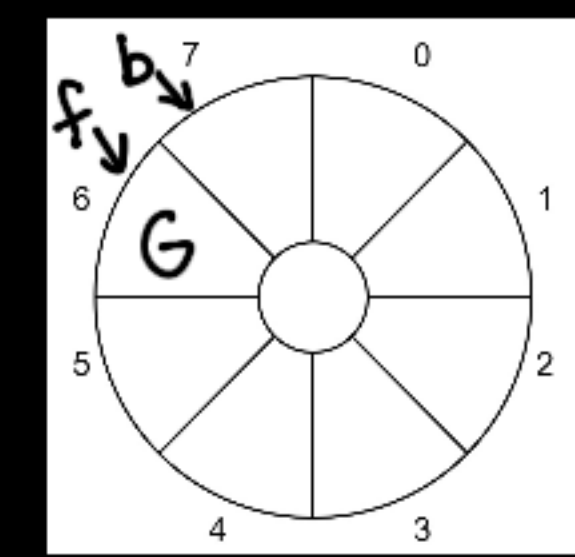


1) enqueue(D)
2) enqueue(E)
This is the rightward shift issue.

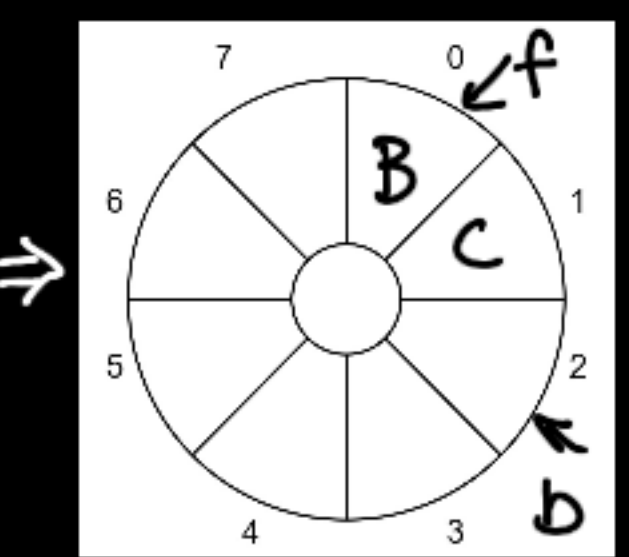
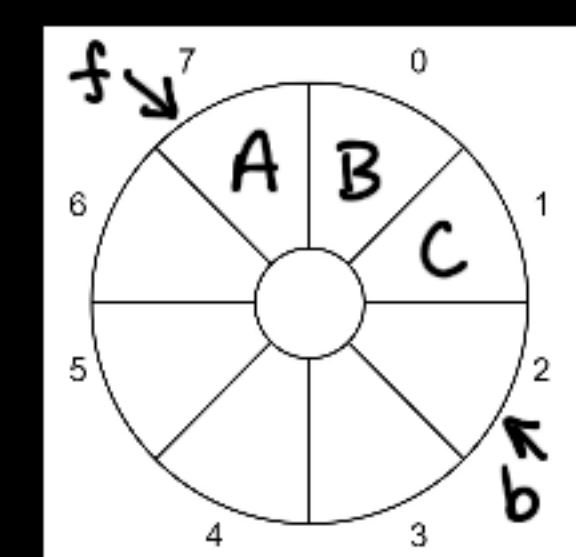
④ Circular Queue



What happens? enqueue(H)
 $b = (b + 1) \% \text{size}$



What happens? dequeue
 $f = (f + 1) \% \text{size}$



⑤ Detective West processes the criminals the Flash catches. Each criminal arrives at a time and takes a given amount of time to process. Criminals are processed in first-come first-serve order. Given the following table of arrivals and process times, show what happens at each step.

<u>C</u>	<u>Arrive</u>	<u>Requires</u>	<u>C</u>	<u>Arr.</u>	<u>Req</u>	<u>Begin</u>	<u>End</u>
A	5	5	A	5	5	5	10
B	7	9	B	7	9	10	19
C	8	4	C	8	4	19	23
D	23	6	D	23	6	23	29
E	30	5	E	30	5	30	35
F	33	4	F	33	4	35	39
G	38	6	G	38	6	39	45

Event Time	<u>A</u>	<u>5</u>	Event Time
	<u>AB</u>	<u>7</u>	
	<u>ABC</u>	<u>8</u>	
	<u>BC</u>	<u>10</u>	
	<u>C</u>	<u>19</u>	
	<u>D</u>	<u>23</u>	
		<u>29</u>	
	<u>E</u>	<u>30</u>	
	<u>EF</u>	<u>33</u>	
	<u>F</u>	<u>35</u>	
	<u>FG</u>	<u>38</u>	
	<u>G</u>	<u>39</u>	
		<u>45</u>	